Week 7 Report

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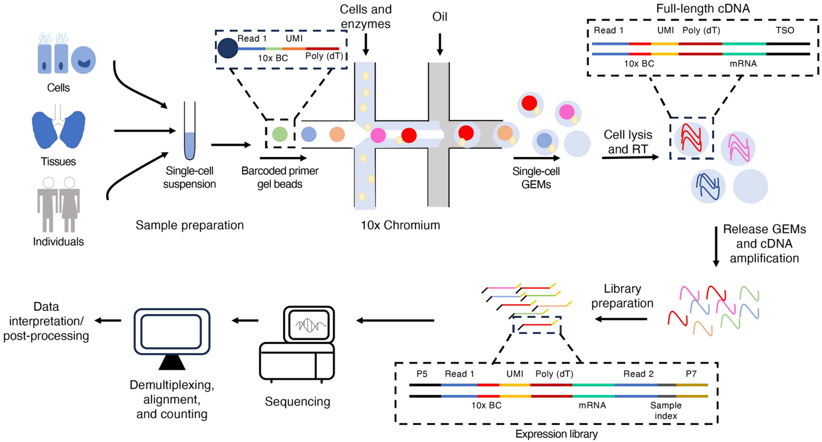
October 31, 2025

1. Single Cell Workflow

2. Bulk COAD GSEA
   - Different sorting choose : T-value vs logFC
   - Random choose and Vote


Cells
Tissues
Individuals
Single-cell suspension
Read 1
UMI
10x BC
Poly (dT)
Barcoded primer gel beads
Sample preparation
Cells and enzymes
Oil
10x Chromium
Single-cell GEMs
Full-length cDNA
Read 1
UMI
Poly (dT)
TSO
10x BC
mRNA
Cell lysis and RT
Release GEMs and cDNA amplification
Library preparation
P5
Read 1
UMI
Poly (dT)
Read 2
P7
10x BC
mRNA
Sample index
Expression library
Sequencing
Demultiplexing, alignment, and counting
Data interpretation/ post-processing

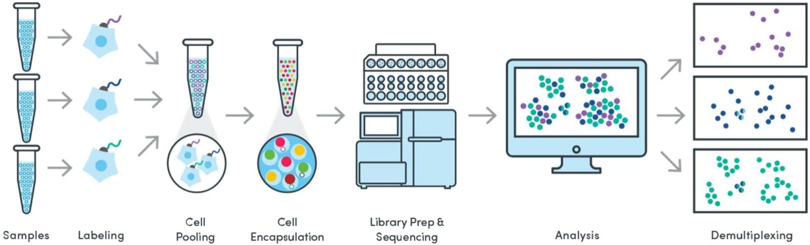
Barcode

1. Read 1
2. Cell Barcode
3. UMI
4. Poly(dt)

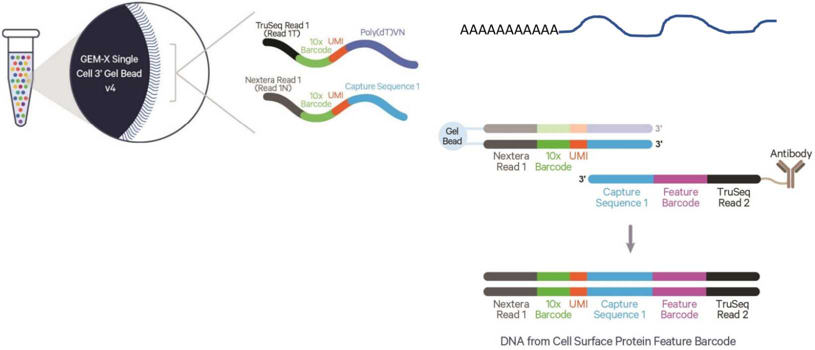
Feature Barcode

1. Antibody Capture (cell surface protein measurement): A method to label cell surface proteins using a specific protein binding molecule, such as an antibody conjugated to a Feature Barcode oligonucleotide.
2. CRISPR Guide Capture: A method to analyze gene expression changes caused by the presence of CRISPR perturbations. The assay captures the full transcriptome and transfected guide RNAs from the same cell to correlate changes in the transcriptome to CRISPR perturbations.
3. 3' Cell Multiplexing: A method to label a given cell (or nuclei) sample with a molecular tag and subsequently pooling this sample with other labeled samples.

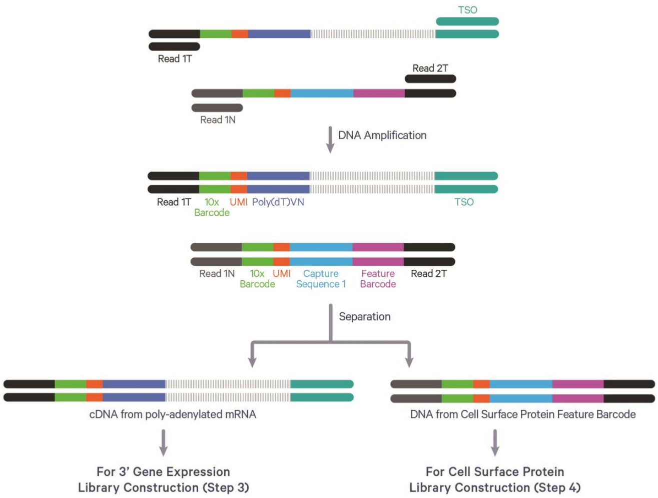
Feature Barcode Technology



Gene Expression & Feature Barcode

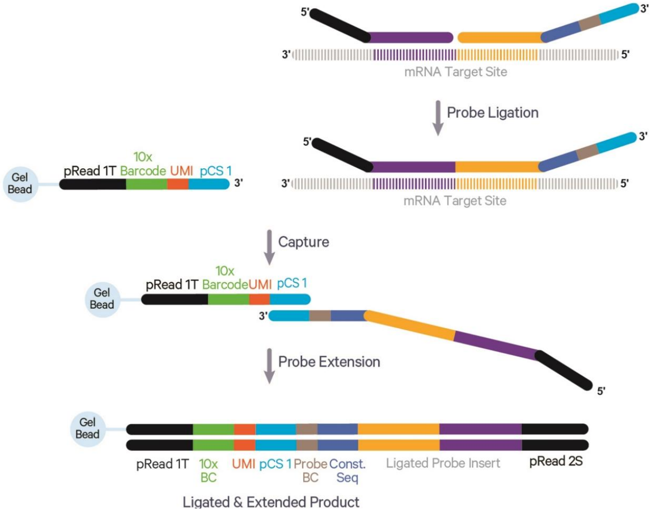


Gene Expression & Feature Barcode



Gene Expression & Feature Barcode

Flex Gene Expression & Feature Barcode



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GSEA

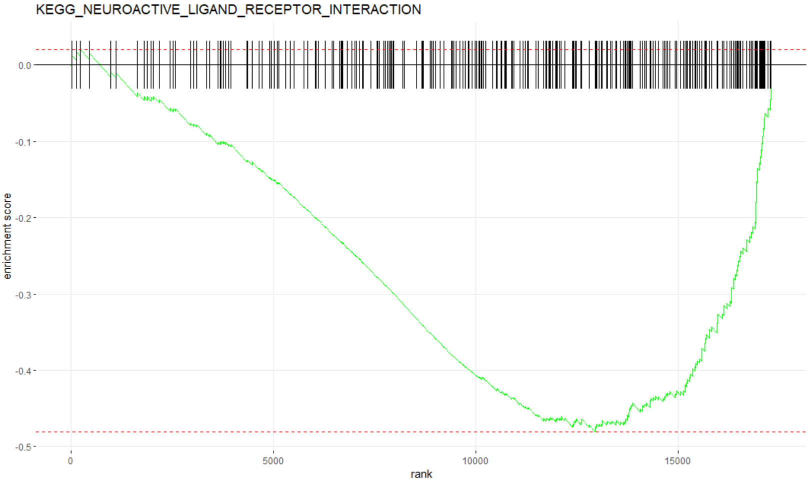
Use **KEGG gene sets**

Number of genes used in GSEA: 17313

Compare **logFC** and **t-value** sorting (Tumor vs Normal).

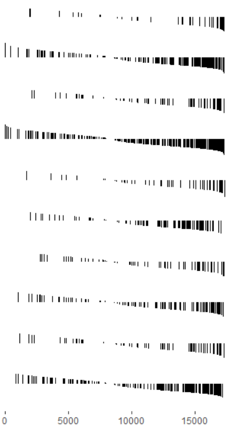
Sorting pathway by

- min padj
- max NES

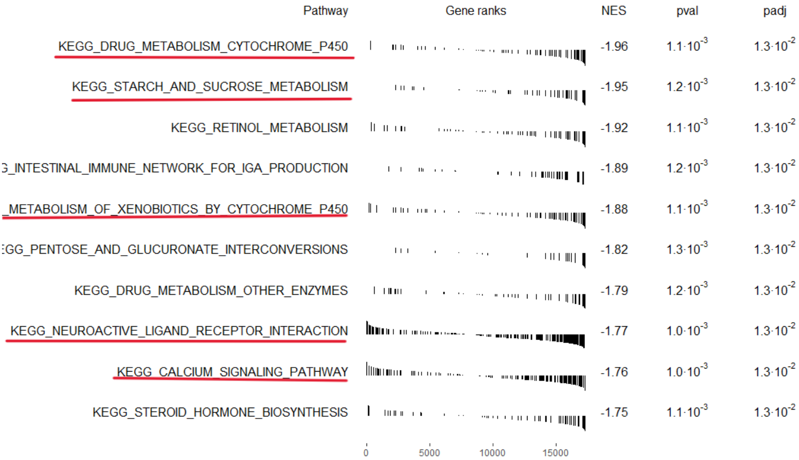

KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERACTION
enrichment score
0.0
-0.1
-0.2
-0.3
-0.4
-0.5
0
5000
10000
15000
rank

Top 10 Pathways by t-value

Pathway	Gene ranks	NES	pval	padj
KEGG_FATTY_ACID_METABOLISM		-2.25	$1.6 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
KEGG_CALCIUM_SIGNALING_PATHWAY		-2.22	$1.3 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
KEGG_DRUG_METABOLISM_CYTOCHROME_P450		-2.16	$1.5 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERACTION		-2.10	$1.2 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
KEGG_STARCH_AND_SUCROSE_METABOLISM		-2.09	$1.5 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
KEGG_PARKINSONS_DISEASE		-2.08	$1.5 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
KEGG_LONG_TERM_DEPRESSION		-2.06	$1.5 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
KEGG_VASCULAR_SMOOTH_MUSCLE_CONTRACTION		-2.04	$1.5 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
_METABOLISM_OF_XENOBIOTICS_BY_CYTOCHROME_P450		-2.00	$1.5 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$
KEGG_CHEMOKINE_SIGNALING_PATHWAY		-1.98	$1.3 \cdot 10^{-3}$	$8.3 \cdot 10^{-3}$



Top 10 Pathways by logFC



Top Pathways

	T-value		logFC	
	padj < 0.05	padj < 0.01	padj < 0.05	padj < 0.01
NES > 0	20	16	10	0
NES < 0	65	44	34	0
Total	**85**	**60**	**44**	**0**

Table 1: Summary of enriched pathways (T-value vs logFC ranking).

he number of pathways intersecting between T-value and logFC is $\mathbf{36}$.

Top 10 Pathways by logFC

Pathway	size	T-value leadingedge Count	logFC leadingedge Count	Common Count
1	176	84	89	76
2	71	30	35	29
3	270	116	98	95
4	48	21	22	19
5	69	28	34	27

1 = KEGG_CALCIUM_SIGNALING_PATHWAY
2 = KEGG_DRUG_METABOLISM_CYTOCHROME_P450
3 = KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERACTION
4 = KEGG_STARCH_AND_SUCROSE_METABOLISM
5 = KEGG_METABOLISM_OF_XENOBIOTICS_BY_CYTOCHROME_P450

Conclusion : logFC vs t-value

Comparison Metric	
padj	**t-value** < logFC
Number of pathways (padj < 0.05)	**t-value** > logFC
|NES|	**t-value** > logFC
Leading edge gene count	t-value < **logFC**

We choose t-value sorting and

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Idea

I have no idea how to choose the pathway.

Initially, we consider the **K-fold** strategy. However, because pathways themselves do not serve as evaluation functions, we propose combining the results across folds using a voting mechanism, analogous to the approach employed in SVM classification.

For each iteration, we randomly sample $r = \{0.4, 0.6, 0.8, 1\}$ of data from both Normal and Tumor groups. The **pathway with the highest adjusted p-value (padj)** is determined via a voting mechanism. This process is repeated **100 times**, and pathways appearing in more than 90 iterations are considered significant.

Pathway Counts for r_0.4

Pathway	Count	Pathway	Count	Pathway	Count	Pathway	Count
1	3	2	2	3	1	4	4
5	1	6	9	8	1	9	3
15	1	18	1	21	1	30	1
35	1	37	1	39	1	44	1
45	1	50	1	51	1	62	1
67	1	69	1	71	1	74	1
77	1	81	1	82	1	90	1
91	1	93	4	94	1	95	1
96	1	97	1	98	1	99	1
100	15						

Number of Count $>$ 90 is **26**

Pathway Counts for r_0.6

Pathway	Count	Pathway	Count	Pathway	Count	Pathway	Count
1	5	2	4	3	2	4	10
5	1	7	1	14	1	17	1
19	1	20	2	30	1	32	1
33	1	37	1	44	1	48	1
50	1	55	1	68	1	69	1
71	1	73	2	77	1	83	1
88	1	90	2	91	1	92	1
94	1	95	1	96	1	97	2
98	2	99	3	100	12		

Number of Count $>$ 90 is **24**

Pathway Counts for r_0.8

Pathway	Count	Pathway	Count	Pathway	Count	Pathway	Count
1	2	2	3	4	5	5	2
6	10	7	1	9	1	10	1
13	1	15	1	17	1	23	1
30	2	36	1	38	1	42	2
43	1	53	1	61	1	62	1
68	2	72	1	80	1	81	1
82	1	86	1	89	1	90	1
91	2	92	1	94	1	95	1
96	1	97	4	98	1	100	13

Number of Count $>$ 90 is **24**

Pathway Counts for r_1

Pathway	Count	Pathway	Count	Pathway	Count	Pathway	Count
1	2	2	4	3	14	5	1
6	1	7	2	11	1	17	1
18	1	19	1	26	1	34	1
36	1	40	1	41	1	43	1
49	1	53	1	62	1	65	1
67	1	72	1	76	1	77	1
84	1	86	1	89	2	91	1
92	3	93	1	94	2	95	1
97	1	99	3	100	13		

Number of Count $>$ 90 is **25**

Intersection Pathways (count > 90)

KEGG_FATTY_ACID_METABOLISM
KEGG_CALCIUM_SIGNALING_PATHWAY
KEGG_DRUG_METABOLISM_CYTOCHROME_P450
KEGG_PARKINSONS_DISEASE
KEGG_STARCH_AND_SUCROSE_METABOLISM
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERACTION
KEGG_VASCULAR_SMOOTH_MUSCLE_CONTRACTION
KEGG_LONG_TERM_DEPRESSION
KEGG_METABOLISM_OF_XENOBIOTICS_BY_CYTOCHROME_P450
KEGG_GNRH_SIGNALING_PATHWAY
KEGG_INTESTINAL_IMMUNE_NETWORK_FOR_IGA_PRODUCTION
KEGG_CHEMOKINE_SIGNALING_PATHWAY
KEGG_PPAR_SIGNALING_PATHWAY
KEGG_VALINE_LEUCINE_AND_ISOLEUCINE_DEGRADATION
KEGG_LONG_TERM_POTENTIATION
KEGG_CARDIAC_MUSCLE_CONTRACTION
KEGG_PHOSPHATIDYLINOSITOL_SIGNALING_SYSTEM
KEGG_ALZHEIMERS_DISEASE
KEGG_REGULATION_OF_ACTIN_CYTOSKELETON
KEGG_ALDOSTERONE_REGULATED_SODIUM_REABSORPTION

Intersection Pathways (count == 100)

KEGG_FATTY_ACID_METABOLISM
KEGG_CALCIUM_SIGNALING_PATHWAY
KEGG_DRUG_METABOLISM_CYTOCHROME_P450
KEGG_PARKINSONS_DISEASE
KEGG_STARCH_AND_SUCROSE_METABOLISM
KEGG_NEUROACTIVE_LIGAND_RECEPTOR_INTERACTION
KEGG_VASCULAR_SMOOTH_MUSCLE_CONTRACTION
KEGG_LONG_TERM_DEPRESSION
KEGG_METABOLISM_OF_XENOBIOTICS_BY_CYTOCHROME_P450
KEGG_GNRH_SIGNALING_PATHWAY
KEGG_CHEMOKINE_SIGNALING_PATHWAY